Problem 1

$$g(x,y) = \frac{(x+y)^2}{x^2+y^2}, \qquad (x,y) \neq (0,0).$$

1. Find and sketch the level curves $g = 0$, $g = 1$, and $g = 2$.
2. Use two of these level curves as paths to determine $\lim_{(x,y)\to(0,0)} g(x,y)$.
3. Now let

$$h(x,y) = \frac{x^2 y}{x^4+y^2}, \qquad (x,y) \neq (0,0).$$

   Show that $h(x,y) \to 0$ as $(x,y) \to (0,0)$ along every straight line through the origin.
4. Show that $\lim_{(x,y)\to(0,0)} h(x,y)$ does not exist.

Problem 2

$$f(x,y) = \begin{cases} \dfrac{xy^3}{x^2+y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

1. Use the definition of partial derivatives to compute $f_x(0,0)$ and $f_y(0,0)$.
2. Show that
$$f_x(0,y) = y \quad \text{for all } y, \qquad f_y(x,0) = 0 \quad \text{for all } x.$$
3. Use part 2 to compute $f_{xy}(0,0)$ and $f_{yx}(0,0)$. Are they equal?
4. For $(x,y) \neq (0,0)$,
$$f_x(x,y) = \frac{y^3(y^2-x^2)}{(x^2+y^2)^2}.$$
Use this formula to show that f_{xy} is not continuous at $(0,0)$. Hence explain why Clairaut's Theorem does not apply.